

Practice: Institute Learning Management System

Class, object, data hiding, encapsulation, association relationship, inheritance relationship

Scenario

A university is building a simple Learning Management System (LMS) to manage students, instructors, courses, and assignments.

The university has many students and instructors. Every student has personal academic information, while every instructor belongs to a department and teaches courses.

Each student has a personal profile that stores additional information such as contact details and address.

Students can enroll in multiple courses during a semester. Each course is taught by a single instructor and contains several assignments that students need to complete.

The university administration wants a system where they can easily manage courses, instructors, students, and assignments while maintaining proper relationships between them.

Your Tasks

Build a system that can perform the following operations.

Student Management

- Add new students.
- Store each student's personal profile.
- Display a student's complete information, including profile details.

Instructor Management

- Add instructors.
- Assign instructors to courses.
- Display the courses taught by an instructor.

Course Management

- Create courses.

- Assign an instructor to each course.
- Allow multiple students to enroll in a course.
- Display all students enrolled in a course.

Assignment Management

- Create multiple assignments for a course.
- Display all assignments for a selected course.

Display Student Info

User will provide student Id/rego no, you will display:

- Student information
 - Personal profile
 - Enrolled courses
 - Instructor for each course
 - Assignments available in each enrolled course
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Sample Scenario

Create data for the following example:

- 2 Instructors
- 5 Students
- A profile for each student
- 3 Courses
- At least 3 Assignments for each course
- Enroll students in different courses

Finally, display:

1. All instructors and the courses they teach.
2. Every course with its instructor, enrolled students, and assignments.
3. Every student's profile along with the courses they have enrolled in.